

Oğuzhan Yetimoğlu

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SUMMARY

Computer Science graduate from the Technical University of Munich with 3+ years of experience in full stack engineering, cloud native systems, ML driven pipelines, and DevOps. I design and build scalable backend and API architectures, distributed and microservice oriented systems, and intuitive user focused interfaces using Python, TypeScript, React, Docker, Kubernetes, and cloud platforms. I enjoy tackling complex engineering problems, optimizing data and ML workflows, and delivering reliable end to end solutions in production environments.

WORK EXPERIENCE

Siemens (Munich) - Software Engineer (Working Student)

06/2024 – 09/2025

- Modernized a cloud based security analytics platform using Java, Docker, Azure, and Apache NiFi, improving stability and dataflow reliability across ingestion and monitoring workloads.
- Built an LLM powered network intrusion detection module (Python, embeddings, anomaly detection) that strengthened detection quality in distributed data pipelines.
- Optimized containerized microservices to reduce resource consumption and improve runtime efficiency, enabling more predictable workloads with lower hardware footprint in Azure based environments.
- Improved microservice build consistency by restructuring Dockerfile stages, cleaning dependency layers, and ensuring reproducible container environments across Azure deployments.

Ludwig Maximilian University (Munich) - Graduate Research Assistant

10/2023 – 06/2024

- Processed large scale brain MRI neuroimaging datasets using ML and statistical modeling techniques to classify disease presence and subtype patterns.
- Parallelized both preprocessing and processing pipelines on HPC clusters, significantly accelerating feature extraction, data transformation, and model execution workloads.
- Improved research workflow reliability by automating ETL steps, standardizing feature extraction procedures, and optimizing job scheduling to reduce idle compute cycles.

Aimyca GmbH (Munich) - Software Engineer (Working Student)

03/2022 – 03/2023

- Delivered full stack features using Python, Node.js, React, Vue, TypeScript, and REST APIs in an agile SaaS environment.
- Built user centric interfaces with clean component architectures and optimized interaction flows, improving overall frontend usability.
- Developed Docker based CI/CD pipelines on GCP to stabilize deployments and improve system reliability.
- Enhanced conversational AI features through LLM fine tuning, improved prompt structures, and refined extraction logic.
- Collaborated directly with founders on architecture, technical design, and product level prioritization.

NETLAB, Boğaziçi University - Graduate Research Assistant / Software Engineer

02/2020 – 12/2021

- Implemented simulation algorithms and C++/Python testbeds for TÜBİTAK funded molecular communication research, improving experiment reproducibility.
- Built embedded integrations with Raspberry Pi and Arduino, stabilizing hardware based data acquisition pipelines.

Idea Teknoloji (Istanbul) - NLP / ML Intern

06/2019 – 07/2019

- Developed NLP and ML components using Python, Pandas, and statistical modeling, building classification prototypes and exploratory pipelines to support model evaluation in an R&D environment.

CTech Bilişim Teknolojileri (Istanbul) - Cybersecurity ML Intern

01/2019 – 02/2019

- Implemented deep learning based attack classification models and improved preprocessing pipelines for cybersecurity datasets.

VeriPark (Istanbul) - Database Engineering Intern

07/2018 – 09/2018

- Developed and optimized SQL modules for CRM oriented backend systems, improving data consistency and internal tooling workflows.

EDUCATION

Technical University of Munich - M.Sc. Informatics (Grade: 1.6/1)

09/2021 – 09/2025

Boğaziçi University - M.Sc. Computer Engineering (Grade: 4.0/4)

09/2019 – 09/2022

Boğaziçi University - B.Sc. Computer Engineering

09/2014 – 07/2019

PROJECTS

LoRemaster — Mobile Quiz App for Fantasy & Sci-Fi Fandoms

2026

- Architected a cross-platform mobile quiz app targeting fantasy and sci-fi fandoms (LOTR, Dune, Harry Potter, Warhammer, Star Wars, GoT, Marvel) with daily curated quizzes and an endless mode.
- Built a full-stack monorepo with React Native (Expo), Next.js, TypeScript, Zustand, and Firebase (Firestore, Auth, Cloud Functions) for scheduled jobs, event-driven triggers, and server-side validation.
- Engineered an LLM-powered question generation pipeline with LiteLLM, orchestrating multi-provider models through prompt engineering, automated validation, and provider fallback for reliability and cost optimization.
- Developed a Next.js admin panel with full CRUD over realms, users, and AI-generated questions, plus runtime configuration of daily quiz size, content, and publishing workflow.
- Designed a token-based theming system where each universe swaps four CSS variables and an ornament glyph, preserving brand identity without compromising answer-state feedback accessibility.

Elysio.ai — AI Driven Virtual Fashion Prototype

01/2024 – 04/2024

- Built a full stack generative AI prototype using Python (FastAPI), TypeScript, React (Next.js), and MUI.
- Integrated Stable Diffusion and Replicate API for personalized image generation, pose controlled synthesis, and NeRF style 3D reconstruction.
- Designed modular cloud deployment on AWS using Docker, CI/CD, and environment separation for scalable inference workloads.
- Collaborated with the business concept team to translate product requirements into technical generative AI workflows.

Algorithmic Textual Comparison Platform (TUM IDP)

12/2023 – 03/2024

- Developed a full stack platform for financial report comparison with NLP based semantic similarity (TF IDF, BERT embeddings, cosine similarity).
- Built the frontend using Vue.js, Nuxt.js, and Tailwind CSS with dynamic interfaces for upload, search, and comparison visualization.
- Implemented backend APIs in Python (Django) with PostgreSQL, authentication, and document management.
- Designed the core text comparison pipeline and coordinated task planning and development sprints.

Nanosim — GPU Accelerated Simulation Library

10/2019 – 02/2020

- Implemented a GPU accelerated simulation library for molecular communication using C++, CUDA, and Python.
- Contributed to model validation, debugging, and performance tuning for large scale computational experiments.

SKILLS

Programming and Frameworks	Python, TypeScript, JavaScript, Java, C++, React, Next.js, Vue, Nuxt.js, Node.js, FastAPI, Django, Flask, RESTful APIs
Cloud and DevOps	Docker, Kubernetes, Infrastructure as Code, CI/CD, GitHub Actions, GitLab CI, Linux, AWS, GCP, Azure, container orchestration, microservices, cloud native environments
Data and Machine Learning	Pandas, NumPy, SciPy, scikit-learn, PyTorch, ML, NLP, LLMs, RAG, embeddings, vector databases, semantic search, data pipelines
Databases and Storage	PostgreSQL, MySQL, SQLite, MongoDB
Tools and Engineering Practices	OOP, debugging, testing, API design, performance optimization, monitoring, observability, logging pipelines, automated deployments, Bash, shell scripting
Frontend Technologies	Tailwind CSS, Material UI, component based architecture, responsive UI design

PUBLICATIONS

1 journal and 3 conference papers in signal processing and molecular communication.

LANGUAGES

Turkish (Native) • English (Advanced) • German (Pre Intermediate)